OXPECKER AI

A Unified National E-Health Card, Multilingual Clinical AI Triage & Prescription Intelligence Platform for Bangladesh

Initiative: Bangladesh e-Health Hackathon & Symposium 2026	**Target Domain:** Digital Health, E-Health Card & Triage
Leadership: Ministry of Health and Family Welfare (MOHFW)	**Strategic Partner:** World Health Organization (WHO)
Live Prototype: https://oxpecker.equisaas-bd.com	**Status:** 100% Functional & Cloud Deployed (Azure)
Lead Team: Oxpecker AI Engineering & Clinical Systems Team	**Submission Date:** October 2026

1. Executive Summary

Oxpecker AI is a transformative national healthcare platform designed specifically to address the structural bottlenecks of Bangladesh's health ecosystem: severe hospital overcrowding, unreadable handwritten prescriptions with high out-of-pocket costs, and the complete absence of a unified, interoperable citizen health record system. Fully aligned with the **Smart Bangladesh 2041 Vision** and the **MOHFW-WHO E-Health Strategic Plan**, Oxpecker AI establishes a 4-pillar digital continuum:

1. Smart HealthPass (E-Health Card)	A dynamic, QR-coded digital patient identity card that aggregates blood group, allergies, chronic conditions, and past clinical records - accessible in seconds by any ER doctor via standard mobile camera without expensive dedicated hardware.
2. Multilingual Clinical AI Doctor	An Azure OpenAI-powered conversational assistant that understands natural Bangla, English, and Banglish (e.g. *'amar jor jor lagche'*). Grounded in 150 DGHS & WHO clinical guidelines, it resolves routine mild symptoms at home with safe OTC guidance and automates hospital appointment serials to end chaotic queues.
3. Handwritten Prescription & Price Engine	An advanced multimodal vision system that transcribes messy handwritten doctor prescriptions, extracts dosage regimens (1+0+1), computes 7/15/30-day course costs, and matches high-cost branded medicines with affordable, DGDA-approved generic alternatives - saving citizens 30% to 40% on medicine costs.
4. National Emergency Triage (999 & 16263)	Automated clinical red-flag detection (severe chest pain, respiratory distress, trauma) that triggers high-priority visual alerts and one-touch routing to National Emergency 999 and DGHS Shastho Batayon 16263.

2. Problem Statement & National Healthcare Challenges

Despite dramatic gains in maternal health and life expectancy, Bangladesh faces three acute structural hurdles that choke healthcare delivery:

Challenge	Current Ground Reality in Bangladesh	Oxpecker AI Technological Solution
Hospital Overcrowding & OPD Line Paralysis	Over **65% to 70%** of patients crowding Upazila Health Complexes (UHC) and tertiary Medical Colleges suffer from self-limiting or mild illnesses (seasonal cold, mild gastritis, tension headache). Doctors get less than 90 seconds per patient, causing diagnostic exhaustion.	**24/7 AI Clinical Triage:** Handles pre-hospital assessment, home first-aid, safe OTC care (Paracetamol/Antacids), and issues automated booking tokens for patients genuinely requiring in-person physician visits.
Unreadable Prescriptions & Out-of-Pocket Expense	More than **80%** of prescriptions are handwritten, leading to frequent pharmacy dispensing errors. Furthermore, out-of-pocket health expenditure in Bangladesh is an alarming **67%**, largely driven by costly branded medications.	**Multimodal Vision & Generic Comparator:** Decodes illegible doctor handwriting into clean structured data and provides instant price comparisons with DGDA-certified generic alternatives, saving families BDT ■■■-BDT ■■■■ per course.
Fragmented Paper Records & Emergency Blindness	Patients carry paper slips that tear, fade, or get misplaced. In an accident, unconscious state, or emergency, doctors have zero visibility into drug allergies (e.g. Penicillin) or chronic conditions (Diabetes/Hypertension).	**Dynamic HealthPass (E-Health Card):** A unified digital passport with encrypted QR verification, instantly pulling critical clinical dossiers in any hospital across the country.

3. Core Technological Capabilities (Working Live in Prototype)

Unlike speculative concept pitches, **Oxpecker AI is an already built, production-tested platform** hosted on Microsoft Azure cloud infrastructure. Its key architectural modules include:

A. Smart HealthPass (Unified E-Health Card Architecture)

Every registered citizen receives a unique **Oxpecker Health ID (OHID)** accompanied by an encrypted, dynamically rendering QR code. In an emergency, paramedics or attending hospital physicians simply point their smartphone camera at the physical card or digital screen to reveal: blood group, critical drug allergies, ongoing medication regimens, and emergency guardian contact. No specialized hospital scanners or expensive card readers are necessary.

B. Multimodal Handwritten Prescription Reader & Price Savings

Powered by Azure OpenAI gpt-5-mini Multimodal Vision and Azure Computer Vision Read OCR, the engine transcribes cursive physician handwriting with clinically calibrated anti-hallucination guardrails. The system extracts: Doctor Name & BMDC Reg, Patient Demographics, Medicine Names, Strengths, and Dosage schedules (e.g. '■+■+■ ■■■■■■■ ■■ ■■■■■ ■■■■■■'). It instantly queries our Bangladesh Pharma Knowledgebase to calculate 7-day, 15-day, and 30-day course costs and displays equivalent generic alternatives (e.g., *Sergel 20mg @ BDT ■ vs Finix 20mg @ BDT ■ - Saving BDT ■■ per strip*).

C. Clinical AI Triage Engine Grounded in DGHS Protocols

Oxpecker AI embeds a verified dataset of **150 Bangladesh-prevalent diseases** and **9,600+ clinical cases** mapped to DGHS/WHO protocols. If a patient describes symptoms in colloquial Banglish or Bengali, the system assesses symptom severity and either suggests safe home care or directly renders an interactive specialist recommendation card (e.g., Cardiology, Neurology, Gynaecology) to book appointments at nearby hospitals.

D. Strict Non-Medical Guardrails & Privacy Assurance

To ensure uncompromising clinical safety, the system incorporates strict domain guardrails: non-medical queries (coding, trivia, politics) are immediately refused with a professional physician disclaimer. Furthermore, uploaded non-medical photos (such as food, animals, or vehicles) are explicitly rejected to prevent hallucinated diagnoses.

4. Technical Architecture & Security Specifications

Layer	Technologies Used	Enterprise Role & Strategic Capability
Frontend & UI	Next.js 15 (App Router), React 19, Tailwind CSS v4, Framer Motion, Hind Siliguri Typography	Ultra-fast, mobile-first responsive PWA accessible on low-cost Android phones and 3G/4G rural networks.
Multimodal AI	Microsoft Azure OpenAI (gpt-5-mini GlobalStandard S0), Azure Computer Vision Read OCR	Sub-second token streaming, multimodal image analysis, handwriting transcription, and Bangla natural language understanding.
Backend & Engine	FastAPI (Python 3.13), Uvicorn, Azure App Services (Linux containerized)	High-concurrency clinical engine matching patient symptoms against 150 DGHS-verified disease protocols.
Data & HealthPass	PostgreSQL with Row-Level Security (RLS), Supabase Auth, Azure Blob Storage (Medical Records)	HIPAA/GDPR-compliant encrypted health record storage, QR generation, and real-time appointment sync.
National Links	National Emergency (999), Shastho Batayon (16263) Tele-doctor API	Seamless one-tap escalation for life-threatening acute medical emergencies.

5. National Phased Rollout Roadmap

Oxpecker AI is designed for seamless integration with existing government healthcare infrastructure (DHIS2, Community Clinics, and DGHS MIS):

Phase	Timeline	Milestones & Nationwide Deliverables
Phase 1: Pilot & Validation	Months 1 - 3	Deploy in **10 Upazila Health Complexes (UHC)** and **50 Community Clinics** in Dhaka and Rajshahi divisions. Validate prescription OCR against 10,000 real-world prescriptions; gather user feedback from 25,000 rural patients.
Phase 2: District Expansion	Months 4 - 8	Scale across **16 District Sadar Hospitals** and integrate with DGHS MIS patient database. Roll out digital appointment token kiosks to slash OPD queue wait times by 40%.

Phase 3: Nationwide Rollout	Months 9 - 18	Universal distribution of the **E-Health Card (HealthPass)** across all 64 districts in partnership with Union Digital Centers (UDC). Onboard 14,000+ Community Clinics to serve 200 million citizens nationwide.

6. Social Impact & Economic Sustainability

Measurable National Impact:
• **30%-40% Out-of-Pocket Expense Reduction:** Generic alternatives and cost visibility empower poor patients to avoid overpriced branded drugs.
• **45% Reduction in Hospital Wait Times:** Digital triage and appointment tokens eliminate the traditional 5-hour wait in humid hospital corridors.
• **Zero Hardware Infrastructure Burden:** Built as a web application utilizing existing smartphones and standard printers for QR health cards.
• **Sustainability Framework:** 100% Free for citizens and rural Community Clinics under a Public-Private Partnership (PPP) model with MOHFW. Sustained through enterprise B2B modules for private diagnostic centers and telemedicine providers.

7. Team Credentials & Prototype Verification

The Oxpecker AI team brings together cross-functional expertise in full-stack cloud engineering, artificial intelligence, and public health informatics:

Team Role	Key Responsibilities & Contribution
Team Leader & Lead AI Developer	Architected end-to-end Azure OpenAI gpt-5-mini multimodal vision pipeline, Next.js 15 frontend, and DGHS clinical triage engine.
Product Pitch & Presentation Lead	Leads health policy alignment, WHO SDG metric validation, stakeholder coordination, and live symposium demonstration.
Clinical Data & Systems Engineer	Maintains 150-disease clinical dataset, DGDA generic pharmaceutical mappings, and HIPAA-compliant PostgreSQL architecture.

8. Conclusion & Readiness for National Deployment

Oxpecker AI is not an abstract concept or theoretical wireframe - it is a live, working, and deployable digital health ecosystem ready to support the Ministry of Health & Family Welfare (MOHFW) in its mission to deliver Universal Health Coverage to Bangladesh. We look forward to presenting the live working system at the **Grand Finale & National Symposium (December 3-5, 2026)**.

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Platform: https://oxpecker.equisaas-bd.com

Verified Working Prototype:
[X] Live Azure OpenAI gpt-5-mini
[X] Live HealthPass Smart Card
[X] Live Prescription OCR & Price Engine